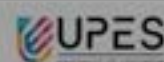


Name: **PRATHAM NA**
Enrolment No: **590032424**



UPES
Assignment 1

Programme Name : B.Tech SoCS/SoAE
Course Name : Advanced Engineering Mathematics I
Course Code : AEM1001
Date of Submission: 02/10/2026

Semester : I
Max. Marks : 20
No. of page(s) : 01

S.No.		Marks	CO
1	Discuss the geometrical interpretation of Rolle's and Lagrange's Mean Value Theorem.	4	CO1
2	Let $s(t)$ be the total miles that you have driven on your trip down a toll road freeway after time t hours. Assume also that $s(1/3) = 10$ miles and $s(1.2) = 82$ miles. At a checkpoint along the toll road, you admit these facts to a highway patrol officer, who is familiar with the <i>Mean Value Theorem</i> . The officer makes a few quick calculations, smiles, then politely prepares to write you a traffic ticket. Explain why.	4	CO1
3	A delivery of 480 cubic meters of gravel is to be made to a landfill. The trucker plans to purchase an open-top box in which to transport the gravel in numerous trips. The total cost to the trucker is the cost of the box plus Rs. 80 per trip. The box must have height 2 meters, but the trucker can choose the length and width. The cost of the box is Rs. 100 per square meter for the ends, Rs. 50 per square meter for the sides and Rs. 200 per square meter for the bottom. Notice the tradeoff: A smaller box is cheaper to buy but requires more trips. What size box should the trucker buy to minimize the total cost?	4	CO1
4	A cantilever beam of length $L = 1\text{m}$ is subjected to a distributed transverse load $w(x)$ (measured in N/m) along its length, where x is the distance from the fixed end $x = 0$ to the free end $x = 1$. Based on experimental strain-gauge measurements, engineers propose the load distribution model: $w(x) = Cx^3(1 - x^2)^4 N/m$ Physical condition: The total load applied to the beam is measured to be exactly $\int_0^1 w(x)dx = 1\text{ N} \quad (\text{Calibration Condition})$ Once C is known, the engineers require the location of the resultant force; the point $\bar{x}$ along the beam at which the entire distributed load could be replaced by a single equivalent point load, given by the centroid of the load distribution: $\bar{x} = \int_0^1 x w(x)dx \quad (\text{in } m \text{ from the fixed end})$ Knowing $\bar{x}$ is essential for computing the maximum bending moment at the fixed end, which governs the beam's structural safety. Using the Beta and Gamma functions, determine the constant C and the centroid $\bar{x}$ of the load distribution.	4	CO2
5	Charge is distributed over the triangular region D given below so that the charge density at (x, y) is $\sigma(x, y) = xy$, measured in coulombs per square meter. Find the total charge.	4	CO2

Assignment-1

Advanced Engineering Mathematics I

Name - Pratham

SAP ID - 590032424

Batch - 20

Course code - AEM1001

Question-1 (a)

Answer-1 Rolle's theorem

If a function $f(x)$ is

- Continuous in the closed interval $[a, b]$
- Differentiable in the open interval (a, b)
- $f(a) = f(b)$

Then there is at least one $c \in (a, b)$ such that $f'(c) = 0$

Geometrical Interpretation:

Let $y = f(x)$ represents a curve with

$A[a, f(a)]$ & $B[b, f(b)]$ as end points $C[c, f(c)]$ be any point between A & B

Let $f'(c)$ = Slope of tangent at point C .

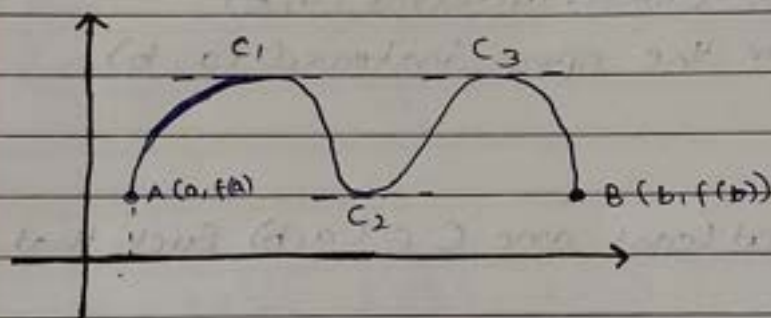
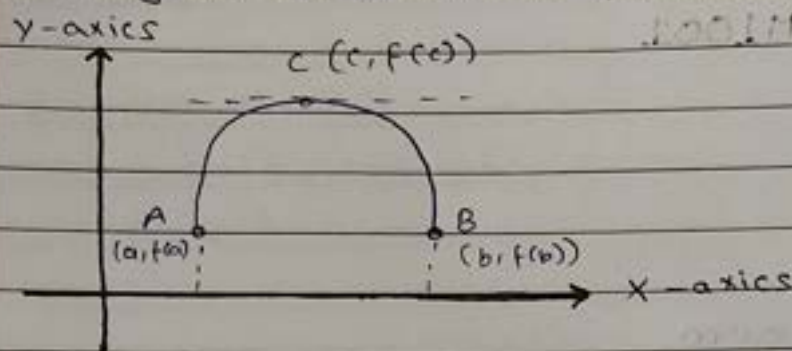
If

- Curve is continuous on $A[a, f(a)]$ and $B[b, f(b)]$ at also at every point between A & B

(ii) Posses unique tangent at every point between A and B

(iii) Ordinate of points A and B are Same i.e., $f(a) = f(b)$

Then there exists at least one point between A and B, tangent at which is parallel to x-axis.



Question 2 (b)

Lagrange's Mean Value Theorem:

If a function $f(x)$ is

- i Continuous in $[a, b]$
- ii Differentiable in (a, b)

Then there at least one $c \in (a, b)$ Such that

$$\frac{f(b) - f(a)}{b - a} = f'(c)$$

Geometrical Interpretation

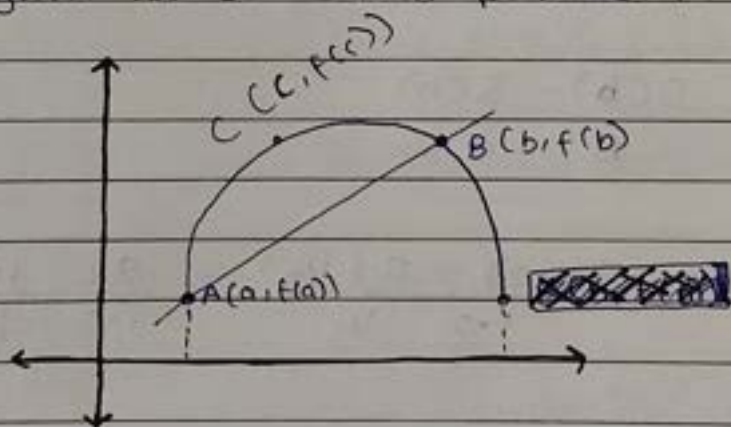
Let $y = f(x)$ represents a Curve with $A(a, f(a))$ and $B(b, f(b))$ as end point and $C(c, f(c))$ be any point between A & B . Let $f'(c) = \text{Slope of the tangent at point } C$.

$$\frac{f(b) - f(a)}{b - a} = \text{Slope of the } \overset{\text{Chord } AB}{\text{tangent at point } C}$$

Then if

- (i) Curve is continuous at A, B & every point A & B
- (ii) Posses unique tangent at every point b/w A & B

Then there at least one point C between A & B tangent at which is parallel to Chord AB .



Question-2

Answer-2 The question gives

$S(t)$ → The function gives function of ~~displacement~~ variable of time.

Given $S(1/3) = 10$ miles and $S(1.2) = 82$ miles
we need to tell why do the ticket was charged on them?

as we know $S'(t)$ → tell the Speed of car at time t .

So by Applying Lagrange's Mean Value Theorem.

we can say

$$S'(c) = \frac{S(b) - S(a)}{b - a}$$

$$\text{So } S'(c) = \frac{S(1.2) - S(1/3)}{1.2 - 1/3} = \frac{82 - 10}{1.2 - \frac{1}{3}} = \frac{72}{1.2 - 0.33}$$

$$= \frac{72}{0.86} = 83.08 \text{ miles/hour}$$

Therefore, the driver must have been travelling at approximately 83 mph at some point during the trip.

If the Speed limit on the highway or road was 80 mph then the driver exceeded the Speed limit and was charged with ticket by police officer.

Question-3

Answer-3

The question states that 480 m^3 of gravel must be transported. The box has a fixed height of 2 m , while length and width can be chosen.

Let $x = \text{length}$

$y = \text{width}$

$2 = \text{Height}$

Solution

The volume of the box $(V) = x \times y \times 2 = 2xy \text{ m}^3$
for one trip

$$V = 2xy$$

There fore number of trip required is:-

$$\frac{480}{2xy} = \frac{240}{xy}$$

Cost of box:

The box is open from the top

There are two ends each of area

$$2y$$

So total end area is

$$4y$$

Cost

$$100(4y) = 400y$$

There are two sides each of area $2x$

So total side area: $4x$

$$\text{Cost} : 50(4x) = 200x$$

Bottom area: xy

$$\text{Cost} : 200xy$$

Therefore total cost of the box is:

$$C_{\text{box}} = 200x + 400y + 200xy$$

Cost of trip

Each trip cost ₹80

$$\text{So total cost of trips} = 80 \times \frac{240}{xy} = \frac{19200}{xy}$$

Hence total cost is

$$C = 200x + 400y + 200xy + \frac{19200}{xy}$$

Minimize the cost:

$$C(x, y) = 200x + 400y + 200xy + \frac{19200}{xy}$$

for the minimum, set the partial derivatives equal to zero

$$\frac{\partial C}{\partial x} = 200 + 200y - \frac{19200}{x^2y} = 0$$

$$\frac{\partial C}{\partial y} = 400 + 200x - \frac{19200}{x^2y} = 0$$

Solving further

$$x^2y(1+y) = 96 \quad \text{--- ①}$$

$$xy^2(2+x) = 96 \quad \text{--- ②}$$

Dividing equation ① and ②

$$\frac{x(1+y)}{y(2+x)} = 1$$

$$\text{Hence } x(1+y) = y(2+x)$$

$$x = 2y$$

Substituting value of $x = 2y$ in equation ①

$$(2y)^2 y (1+y) = 96$$

$$y^3 (1+y) = 24$$

$$y \approx 1.82 \text{ m}$$

Therefore,

$$x = 2y \approx 3.64 \text{ m}$$

Final answer

$$L \approx 3.64 \text{ m}$$

$$W \approx 1.82 \text{ m}$$

$$H = 2 \text{ m}$$

So the trucker should buy a box approximately
 $3.64 \times 1.82 \times 2$

Question-4

Answer-4 The given load distribution is

$$w(x) = Cx^3 (1-x^2)^{3/4} \text{ N/m} \quad 0 \leq x \leq 1$$

The total load is 1 N , So

$$\int_0^1 w(x) \cdot dx = 1$$

$$\text{The Centroid is : } \bar{x} = \frac{\int_0^1 x w(x) \cdot dx}{\int_0^1 w(x) \cdot dx}$$

Solutions:

using the total-load condition:

$$C \int_0^1 x^3 (1-x^2)^{3/4} dx = 1$$

Let $t = x^2$

Then $dt = 2x \cdot dx$

and

$$x^3 dx = x^2 (x dx) = \frac{t}{2} \cdot dt$$

Therefore

$$x^3 dx = x^2 (x dx) = \frac{t}{2} \cdot dt$$

So $C \frac{1}{2} \int_0^1 t (1-t)^{3/4} dt = 1$

using the Beta function

$$B(p, q) = \int_0^1 t^{p-1} (1-t)^{q-1} dt$$

we get $B(p, q) = \int_0^1 t^{p-1} (1-t)^{q-1} \cdot dt,$

$$\frac{C}{2} B\left(2, \frac{7}{4}\right) = 1$$

Now

$$B(p, q) = \frac{\Gamma(p) \Gamma(q)}{\Gamma(p+q)}$$

Hence $\frac{C}{2} \frac{\Gamma(2) \Gamma(7/4)}{\Gamma(15/4)} = 1$

$$2 = 1$$

$$15/4 = \frac{11}{4} + \frac{7}{4} + \frac{7}{4}$$

we obtain

$$B(2, 7/4) = \frac{16}{77}$$

$$\frac{C}{2} \left(\frac{16}{77} \right) = 1$$

$$C = \frac{77}{8}$$

or

$$C = 9.625$$

Since the total load is 1N

$$\bar{x} = \int_0^1 x w(x) dx$$

Again put

$$t = x^2$$

then

$$x^4 dx = \frac{1}{2} t^{3/2} dt$$

So

$$\bar{x} = \frac{C}{2} \int_0^1 t^{3/2} (1-t)^{3/4} dt$$

using Beta function

$$\bar{x} = \frac{C}{2} B\left(\frac{5}{2}, \frac{7}{4}\right)$$

$$\text{Substituting } C = \frac{77}{8}$$

$$\bar{x} = \frac{77}{16} \frac{\sqrt{5/2} \sqrt{7/2}}{\sqrt{17/4}}$$

Simplify

$$\bar{x} \approx 0.735 \text{ m}$$

Final Answer

$$C = \frac{77}{8} = 9.625$$

and

$$\bar{x} = 0.735 \text{ m}$$

So the resultant force acts approximately 0.735 m from the fixed end.

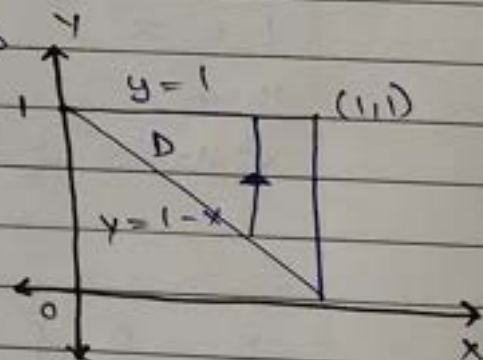
Question-5

Answer-5 The charge density is $\sigma(x,y) = xy$

The triangular region D has vertices $(0,1)$, $(1,1)$, $(1,0)$

and is bounded by

$$y=1 \quad x=1 \quad y=1-x$$



The question ask for the total charge
Solution

The total Charge is

$$Q = \iint_D \sigma(x,y) dA$$

Since

$$\sigma(x, y) = xy,$$

We have

$$Q = \iint_D xy \, dA$$

from the diagram;

$$0 \leq x \leq 1$$

and for each x ,

$$1-x \leq y \leq 1$$

Therefore

$$\int_0^1 \int_{1-x}^1 xy \, dy \, dx$$

$$Q = \int_0^1 x \left[\frac{y^2}{2} \right]_{1-x}^1 dx$$

$$\frac{1}{2} \int_0^1 x [1 - (1-x)^2] dx$$

$$Q = \frac{1}{2} \int_0^1 x (2x - x^2) dx$$

$$= \frac{1}{2} \int_0^1 (2x^2 - x^3) dx$$

$$Q = \frac{1}{2} \left[\frac{2x^3}{3} - \frac{x^4}{4} \right]_0^1$$

$$= \frac{1}{2} \left(\frac{2}{3} - \frac{1}{4} \right)$$

Taking LCM = 12:

$$0 = \frac{1}{2} - \frac{5}{12}$$

$$0 = \frac{5}{24}$$

Final Answer

$$\text{Total Charge} = \frac{5}{24} \text{ Coulombs}$$

or approximately

$$Q \approx 0.2083 \text{ C}$$